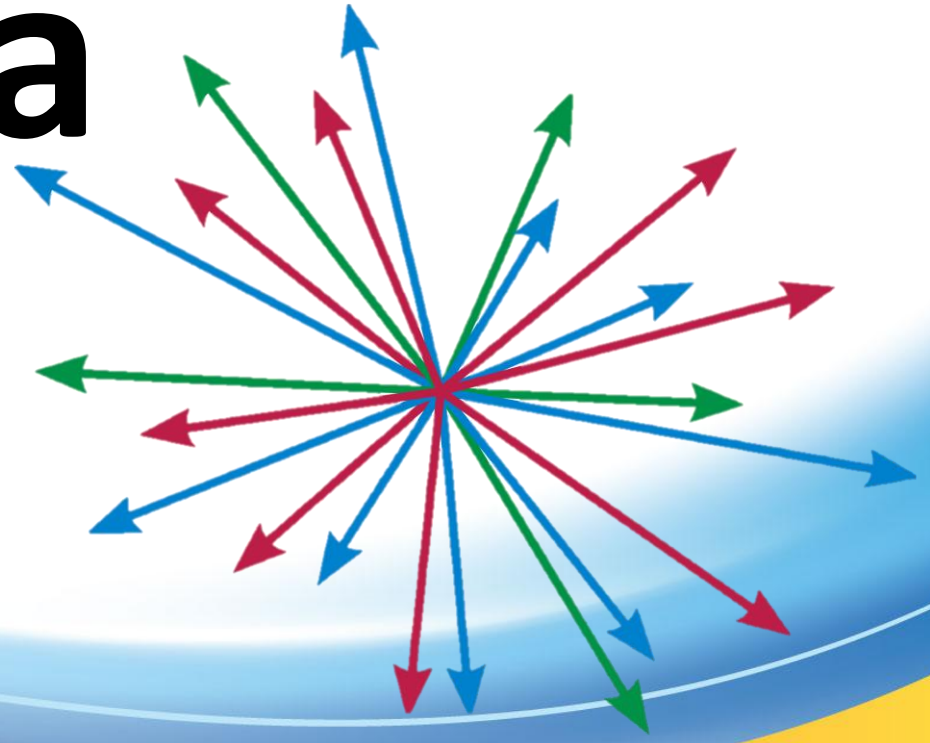
Undergraduate Course in Mathematics

Linear Algebra



Topic: Inner Product

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Inner Product

Abstraction of dot Product.

Inner product

$\vec{u} \cdot \vec{v}$ = Real number / Scalar.

$$(\underline{a}\hat{i} + \underline{b}\hat{j}) \cdot (\underline{c}\hat{i} + \underline{d}\hat{j}) = ac + bd$$

$$(a\hat{i} + b\hat{j}) \cdot (a\hat{i} + b\hat{j}) = a^2 + b^2 \geq 0$$

$$(\vec{u} + \vec{v}) \cdot \vec{w} = \vec{u} \cdot \vec{w} + \vec{v} \cdot \vec{w}.$$

Inner Product over the field $\mathbb{R}$

Let V be a vector space over the field $\mathbb{R}$. An inner product on V is a function that takes each ordered pair $(\vec{u}, \vec{v})$ of elements of V to a number $\langle \vec{u}, \vec{v} \rangle \in \mathbb{R}$ and has the following properties:

Positivity: $\langle \vec{u}, \vec{u} \rangle \geq 0$ for all $\vec{u} \in V$;

Definiteness: $\langle \vec{u}, \vec{u} \rangle = 0$ if and only if $\vec{u} = 0$;

Additivity in first slot: $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$ for all $\vec{u}, \vec{v}, \vec{w} \in V$;

Homogeneity in first slot: $\langle \lambda \vec{u}, \vec{v} \rangle = \lambda \langle \vec{u}, \vec{v} \rangle$ for all $\lambda \in \mathbb{R}$ and for all $\vec{u}, \vec{v} \in V$;

Symmetry: $\langle \vec{u}, \vec{v} \rangle = \langle \vec{v}, \vec{u} \rangle$ for all $\vec{u}, \vec{v} \in V$.

Inner Product over the field $\mathbb{C}$

Let V be a vector space over the field $\mathbb{C}$. An inner product on V is a function that takes each ordered pair $(\vec{u}, \vec{v})$ of elements of V to a number $\langle \vec{u}, \vec{v} \rangle \in \mathbb{C}$ and has the following properties:

Positivity: $\langle \vec{u}, \vec{u} \rangle \geq 0$ for all $\vec{u} \in V$;

Definiteness: $\langle \vec{u}, \vec{u} \rangle = 0$ if and only if $\vec{u} = 0$;

Additivity in first slot: $\langle \vec{u} + \vec{v}, \vec{w} \rangle = \langle \vec{u}, \vec{w} \rangle + \langle \vec{v}, \vec{w} \rangle$ for all $\vec{u}, \vec{v}, \vec{w} \in V$;

Homogeneity in first slot: $\langle \lambda \vec{u}, \vec{v} \rangle = \lambda \langle \vec{u}, \vec{v} \rangle$ for all $\lambda \in \mathbb{C}$ and for all $\vec{u}, \vec{v} \in V$;

Conjugate Symmetry: $\langle \vec{u}, \vec{v} \rangle = \overline{\langle \vec{v}, \vec{u} \rangle}$ for all $\vec{u}, \vec{v} \in V$.

Inner Product in $V = \mathbb{R}^n$

Standard Inner Product in $\mathbb{R}^2$

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = x_1 y_1 + x_2 y_2$$

dot Product

$$\left\langle \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 7 \end{pmatrix} \right\rangle = 8 + 21$$

$$= 29$$

Standard Inner Product in $\mathbb{R}^n$

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} \right\rangle = x_1y_1 + x_2y_2 + \cdots + x_ny_n$$

$$= \sum_{i=1}^{n} x_i y_i$$

Show that

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = 2x_1y_1 + 3x_2y_2$$

Is an inner product in $\mathbb{R}^2$.

YES

Show that

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle = 2x_1y_1 - 3x_2y_2$$

NO

Is not an inner product in $\mathbb{R}^2$.

$$\left\langle \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\rangle = 2 \cdot 1 \cdot 1 - 3 \cdot 2 \cdot 2$$

$$= 2 - 12 = -10 \neq 0$$

Inner Product in $V = \mathbb{C}^n$

Standard Inner Product in $\mathbb{C}^2$

$$\left\langle \begin{pmatrix} z_1 \\ z_2 \end{pmatrix}, \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} \right\rangle = z_1\overline{w_1} + z_2\overline{w_2}$$

$$\left\langle \begin{pmatrix} 2+i \\ 7 \end{pmatrix}, \begin{pmatrix} 2+i \\ 7 \end{pmatrix} \right\rangle \quad (2+i)\cdot\overline{(2+i)} + 7\cdot\overline{7}$$

$$= 4+1+49$$

$$= 54 > 0$$

Standard Inner Product in $\mathbb{C}^n$

$$\left\langle \begin{pmatrix} z_1 \\ z_2 \\ \vdots \\ z_n \end{pmatrix}, \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix} \right\rangle = z_1\overline{w_1} + z_2\overline{w_2} + \cdots + z_n\overline{w_n}$$

Inner Product in Matrix Spaces

Standard Inner Product of 2×3 Real Matrices

$$\left\langle \begin{pmatrix} 4 & 5 & 2 \\ 1 & 0 & 3 \end{pmatrix}, \begin{pmatrix} 2 & 1 & 5 \\ 4 & 2 & 3 \end{pmatrix} \right\rangle$$

$$= \text{Trace} \begin{pmatrix} 4 & 1 \\ 5 & 0 \\ 2 & 3 \end{pmatrix} \cdot \begin{pmatrix} 2 & 1 & 5 \\ 4 & 2 & 3 \end{pmatrix}$$

$$= \text{Trace} \begin{pmatrix} 12 & 6 & 23 \\ 10 & 5 & 25 \\ 16 & 8 & 19 \end{pmatrix} = 36.$$

$$\langle A, B \rangle = \text{Trace}(A^T B)$$

Standard Inner Product of $m \times n$ Real Matrices

$$\langle A, B \rangle = \mathrm{Tr}(A^T B)$$

Inner Product in Function Spaces

Inner Product in the Space of integrable functions

$$\langle f(x), g(x) \rangle = \int_a^b f(x) \cdot g(x)\, dx$$

Complex

$$\langle f(z), g(z) \rangle = \int_a^b f(z) \cdot \overline{g(z)}\, dz$$

$$\langle f(x), g(x) \rangle = \int_0^1 f(x) \cdot g(x)\, dx$$

find the inner product of e^x and x

$$\langle e^x, x \rangle = \int_0^1 e^x \cdot x\, dx = \left[x e^x - e^x \right]_0^1 = e^1 - e^1 - (-e)$$

$$= e.$$

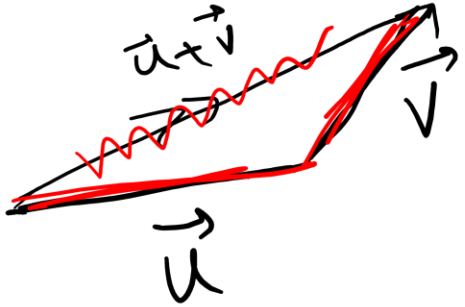
Norm

Abstraction of Length

Norm

Let V be a vector space over the field F. A Norm V is a function that takes each $\vec{u}$ of element of V to a number $\|\vec{u}\| \in \mathbb{R}$ and has the following properties:

Positivity: $\|\vec{u}\| \geq 0$ for all $\vec{u} \in V$;



Definiteness: $\|\vec{u}\| = 0$ if and only if $\vec{u} = 0$;

Triangle Inequality: $\|\vec{u} + \vec{v}\| \leq \|\vec{u}\| + \|\vec{v}\|$ for all $\vec{u}, \vec{v} \in V$;

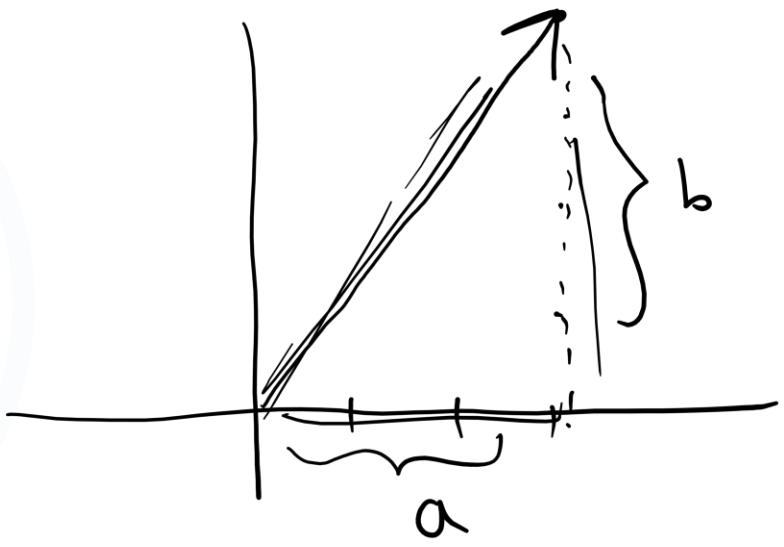
Homogeneity: $\|\lambda\, \vec{u}\| = |\lambda|\, \|\vec{u}\|$ for all $\lambda \in F$ and for all $\vec{u} \in V$;

length from inner product

$$\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}$$

Standard Euclidean Norm in $\mathbb{R}^2$

$$\vec{u} = \begin{pmatrix} a \\ b \end{pmatrix}$$



$$\|\vec{u}\| = \sqrt{a^2+b^2} = \sqrt{\langle \vec{u}, \vec{u} \rangle}$$

$$\langle \vec{u}, \vec{u} \rangle = a^2+b^2$$

Standard Euclidean Norm in $\mathbb{R}^n$

$$\vec{u} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad \|\vec{u}\| = \sqrt{a^2 + b^2 + c^2}$$

$$\vec{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix} \qquad \|\vec{u}\| = \sqrt{u_1^2 + u_2^2 + \cdots + u_n^2}$$

$$\|\sin(x)\| = \sqrt{\langle \sin x, \sin x \rangle}$$

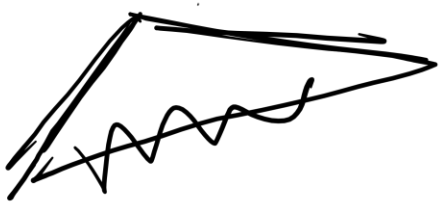
$$= \sqrt{\int_{-\pi}^{\pi} \sin x \cdot \sin x \, dx} = ??$$

Metric

Abstraction of distance

Metric

Let V be a vector space over the field F. An inner product on V is a function $\boldsymbol{d}$ that takes each ordered pair $(\vec{u}, \vec{v})$ of elements of V to a number $\boldsymbol{d}(\vec{u}, \vec{v}) \in \mathbb{R}$ and has the following properties:



Positivity: $\boldsymbol{d}(\vec{u}, \vec{v}) \geq 0$ for all $\vec{u}, \vec{v} \in V$;

Definiteness: $\boldsymbol{d}(\vec{u}, \vec{v}) = 0$ if and only if $\vec{u} = \vec{v}$;

Triangle Inequality: $\boldsymbol{d}(\vec{u}, \vec{w}) \leq \boldsymbol{d}(\vec{u}, \vec{v}) + \boldsymbol{d}(\vec{v}, \vec{w})$ for all $\vec{u}, \vec{v}, \vec{w} \in V$;

Symmetry: $\boldsymbol{d}(\vec{u}, \vec{v}) = \boldsymbol{d}(\vec{v}, \vec{u})$ for all $\vec{u}, \vec{v} \in V$.

$$d(\vec{u}, \vec{v}) = \|\underline{\vec{u}} - \underline{\vec{v}}\|.$$

Standard Euclidean Metric in $\mathbb{R}^2$

$$d\big((a,b),(c,d)\big)$$

$$= \sqrt{(a-c)^2 + (b-d)^2}$$

Standard Euclidean Metric in $\mathbb{R}^n$

$$d\left((x_1, x_2, \ldots, x_n), (y_1, y_2, \ldots, y_n)\right)$$

$$= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

$\langle \vec{u}, \vec{v} \rangle$ inner product

$\|\vec{u}\| = \sqrt{\langle \vec{u}, \vec{u} \rangle}$ norm

$d(\vec{u}, \vec{v}) = \|\vec{u} - \vec{v}\|$ metric

Thank you!